



A preliminary evaluation of the effect of processing temperature on coffee roasting degree assessment

Adriana S. Franca^{a,*}, Leandro S. Oliveira^a, Rafael C.S. Oliveira^a, Pamela C. Mancha Agresti^a, Rodinei Augusti^b

^a Programa de Pós-Graduação em Ciência de Alimentos, UFMG – Av. Antônio Carlos, 6627, 31270-901, Belo Horizonte, MG, Brazil

^b Departamento de Química, UFMG – Av. Antônio Carlos, 6627, 31270-901, Belo Horizonte, MG, Brazil

ARTICLE INFO

Article history:

Received 10 April 2008

Received in revised form 29 September 2008

Accepted 14 December 2008

Available online 25 December 2008

Keywords:

Coffee

Volatiles profile

Roasting degree assessment

SPME/GC–MS

ABSTRACT

The objective of this study was to analyze the volatile profiles of both green and roasted coffee beans, for assessment of roasting degrees under two different processing temperatures. Volatiles extraction and concentration were performed by solid phase micro-extraction (SPME) of the ground coffee headspace and analysis of the volatiles profile was performed by GC–MS. Four SPME fibers were evaluated, with the one that extracted the highest amount of substances (DVB/CAR/PDMS) being selected. Statistical analysis of the data by principal components (PCA/clusters) showed that the volatile components profile provided separation of green and roasted coffees and also separation according to roasting degree and roasting temperature. Results also indicate that color and weight loss measurements alone are not reliable for roasting degree assessment.

© 2008 Elsevier Ltd. All rights reserved.

1. Introduction

The desired aroma and flavor of coffee used for beverage are developed during roasting, a time–temperature dependent process in which the coffee beans undergo a series of reactions leading to several changes in chemical composition. The roasted coffee composition will vary with the processing conditions, being characterized by the degree of roast, which is reflected on the external color of the beans (varying from light to dark brown), the developed flavor, the amount of dry matter loss that occurs and specific changes in chemical composition (Clarke, 1987). In simple terms, roasted coffee is described as presenting a light, medium or dark degree of roast, depending on its color.

The degree of roast is assessed in the industry by measurements of ground beans light reflectance, or just by visual inspection of the color of the beans. However, color-based procedures have been proven ineffective (Purdon and McCamey, 1987; Dutra et al., 2001), since coffee beans roasted to different degrees can present the same average readings in light reflectance measurements. Vasconcelos (2005) showed that good quality coffees submitted to a light roast (30 min roasting at 200 °C) presented the same values for luminosity measurements than low quality coffees submitted to a darker roast (2 h roasting at 200 °C). Furthermore, coffees roasted under the same processing conditions can present different color readings. The study by Franca et al. (2005) showed that both

black and sour beans presented lower luminosity values than non-defective and immature beans roasted under the same conditions. Also, the surface of black beans had the highest values of hue angle after roasting, maintaining a greenish hue while the other types of beans became brown. It was concluded that low quality coffees will attain a lighter roasting degree than good quality ones, under the same processing conditions, with such conclusion being supported by the fact that such beans also presented lower weight loss values. Also, Cristo et al. (2006), in their study of flow regimes in rotating coffee roasters, discussed that smaller beans will tend to migrate to the low velocity core of the bed transverse section. Thus, due to poor mixing (low heat transfer rates) in that region these beans will necessarily roast to a lesser degree, and will probably present a lighter color. In-plant measurements of dry matter loss have been reported to provide a more accurate evaluation of the roasting degree (Clarke, 1987). However, this parameter does not reflect the variations in chemical composition of the coffee beans prior to roasting (Franca and Oliveira, 2008).

A considerable amount of research has been done on the characterization of flavor-related substances in coffee, with over 850 substances being identified in the volatile fraction of roasted coffee (Grosch, 2001; Rocha et al., 2003). Given that the volatile profile changes considerably during roasting and with the original green coffee composition, and also that it presents a direct impact on coffee flavor, it should provide a more reliable measure of the roasting degree. Furthermore, previous studies have demonstrated the feasibility of assessing the degree of roast by multivariate statistical analysis of coffee headspace profiles at varying roasting conditions (Liardon and Ott, 1984) and by evaluating the evolution of the

* Corresponding author. Tel.: +55 31 34093512; fax: +55 31 34433783.

E-mail addresses: adriana@demec.ufmg.br, adrianafranca@pesquisador.cnpq.br (A.S. Franca).

composition of coffee roasting exhaust gases (Dutra et al., 2001). Therefore, the objective of this study was to evaluate the effect of roasting temperature on coffee roasting degree by performing a comparative evaluation of the volatile components profile of both crude coffee and coffees submitted to roasting at similar degrees under two different temperatures.

2. Methodology

Arabica green (crude) coffee samples (2004/2005 crop) were obtained from Santo Antonio Estate Coffee (Minas Gerais, Brazil). Defective coffee beans were hand picked, so that the sampling lot consisted only of good quality coffee. Samples of randomly selected 40 beans were then separated and oven roasted according to the processing conditions specified in Table 1. Roasting degrees were established based on weight loss measurements (percent difference in sample weight before and after roasting) and visual inspection of the external color of the beans, since these are the most common procedures employed by the coffee roasting industry. $L^* a^* b^*$ measurements of ground roasted coffee were also taken to confirm color variations (see luminosity readings displayed in Table 1). Each sample was positioned in the same place in the oven in an effort to ensure uniform roasting conditions. Roasted coffee samples were stored in sealed containers at ambient temperature for a maximum period of 24 h. Just before each analysis, the coffee was ground with an electric coffee grinder (C-mill 5679-01US, Bodum, USA) for 30 s. Commercially available roasted and ground coffee samples were employed for the fiber evaluation tests. Roasting tests were conducted in duplicates.

Four SPME fibers were evaluated for the extraction of volatiles from the roasted and ground coffee headspace: (i) polydimethylsiloxane (PDMS) 100 μ m fiber (model 57324-U, Supelco Inc., USA), (ii) carboxen/polydimethylsiloxane (CAR/PDMS) 75 μ m (model 57324-U, Supelco Inc., USA), (iii) polydimethylsiloxane/divinylbenzene (PDMS/DVB) 65 μ m (model 57324-U, Supelco Inc., USA), and (iv) divinylbenzene/carboxen/polydimethylsiloxane (DVB/CAR/PDMS) 50/30 μ m fiber (model 57348-U, Supelco Inc., USA). Each fiber was conditioned according to manufacturer recommendations prior to use. Ground coffee samples (3 ml) were placed in sealed 5 ml vials (Supelco Inc., USA), and heated for 10 min at 70 °C. The SPME needle was then inserted into the vial, and the fiber exposed to the headspace above the coffee sample for 40 min at 70 °C (Mancha Agresti et al., 2008). After sampling, the fiber was thermally desorbed in the GC injection port for 10 min at 250 °C.

GC analyses were performed using a gas chromatograph (Trace Ultra) coupled to a mass spectrometer (PolarisQ) (ThermoElectron, San Jose, CA), a RTX-5MS column (5% diphenyl, 95% dimethyl polysiloxane) 30 m $\times$ 0.25 mm I.D. (Restec, Ireland), and Helium as carrier gas (1 ml/min). The GC injector was operated at 250 °C in the splitless mode. The ion source (detector) and interface temperatures were 300 °C and 275 °C, respectively. Mass spectra were acquired in the electron impact mode at 70 eV, using a m/z range

of 50–650 and 2 s scan time. Preliminary tests were conducted with variations in the GC oven temperature program based on literature reports (Mayer and Grosch, 2001; Mondello et al., 2004; Ryan et al., 2004; Zambonin et al., 2005), in order to obtain the maximum amount of peaks. Only peaks that presented signal-to-noise ratios above 50 were considered. The optimized conditions obtained for each specific fiber were: (i) PDMS: from 40 °C (5 min) to 100 °C at the rate of 3 °C/min, then to 250 °C (5 min) at 5 °C/min; (ii) CAR/PDMS: from 40 °C (5 min) to 125 °C at the rate of 2.5 °C/min, then to 245 °C (3 min) at 10 °C/min; (iii) PDMS/DVB: from 40 °C (5 min) to 180 °C at the rate of 3 °C/min, then to 250 °C (5 min) at 10 °C/min; and (iv) DVB/CAR/PDMS: from 40 °C (5 min) to 180 °C at the rate of 3 °C/min, then to 250 °C (5 min) at 10 °C/min. Tentative identification of compounds was performed by comparison of the mass spectra with those of the NIST/EPA/NIH Mass Spectral Library, version 2.0 (available in the instrument software), considering a similarity level (RSI) higher than 800.

For principal component analysis (PCA), data matrices were assembled so that each row corresponded to a coffee sample and each column represented the relative peak area of a specific component. Complete linkage method and Euclidean distance were used to generate the dendrograms in the HCA method. Regardless of substance tentative identification all chromatographic peaks with S/N (signal-to-noise ratio) higher than 50 were included in the final data matrix.

3. Results and discussion

Results for weight loss during roasting are shown in Fig. 1, with the dashed lines representing logarithmic model fit (Fig. 1a) and

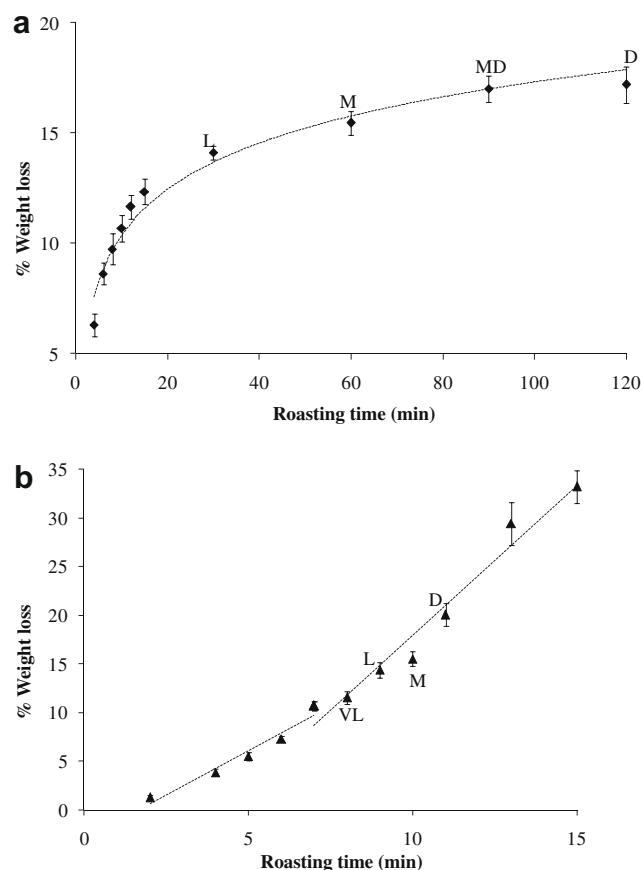


Fig. 1. Weight loss during roasting at: (a) 200 °C and (b) 300 °C. VL = very light roast, L = light roast, M = medium roast, MD = medium/dark roast, D = dark roast.

Table 1
Roasting conditions.

Sample ID	Roasting degree (luminosity range)	Oven temperature (°C)	Roasting time (min)
L	Light (29–31)	200	30
M	Medium (26–27)	200	60
MD	Medium/dark (25–26)	200	90
D	Dark (23–24)	200	120
VL	Very light (32–35)	300	8
L	Light (29–31)	300	9
M	Medium (26–27)	300	10
D	Dark (23–24)	300	11

linear model fit (Fig. 1b). Notice that time scales in Fig. 1a and b are different because coffee roasts faster at higher temperatures. Average mass loss values per roasting degree were 14%, 15%, and 19%, corresponding to light, medium, and dark roasts, respectively. It is noteworthy to mention that roasting temperature presented a significant effect on weight loss behavior. At the lower temperature (200 °C), weight loss occurred faster during the first 20 min, which corresponds to the drying stage during which weight loss is mostly attributed to removal of water and crude coffee volatiles. Also, weight loss behavior was satisfactorily described by a logarithmic fit (dashed line in Fig. 1a). When roasting temperature was increased (300 °C), the weight loss presented a different behavior, with the drying stage occurring at a slower rate than the pyrolysis stage. After the drying stage (0–7 min roasting), there was a significant increase in the weight loss rate that can be attributed to an intensive release of organic compounds, carbon dioxide,

and water resultant from pyrolysis and other roasting reactions. This behavior has been reported by previous studies on coffee roasting (Sivetz and Desrosier, 1979; Dutra et al., 2001; Oliveira et al., 2005). The onset of pyrolysis can be associated with the transition between the two slopes. Weight loss results were employed to establish roasting degree correspondence between different roasting temperatures, since such measurement has been reported to be more reliable than color inspection alone (Clarke, 1987). However, the variation in weight loss behavior shown in this study is an indication that the roasting temperature is affecting the volatiles profile by favoring different reactions and product removal pathways at different processing conditions.

Typical chromatograms obtained for commercial roasted coffee using different SPME fibers are presented in Fig. 2. It was possible to extract and detect a great number of volatile compounds. Based on the aforementioned criteria, total amounts of 80, 107, 128, and

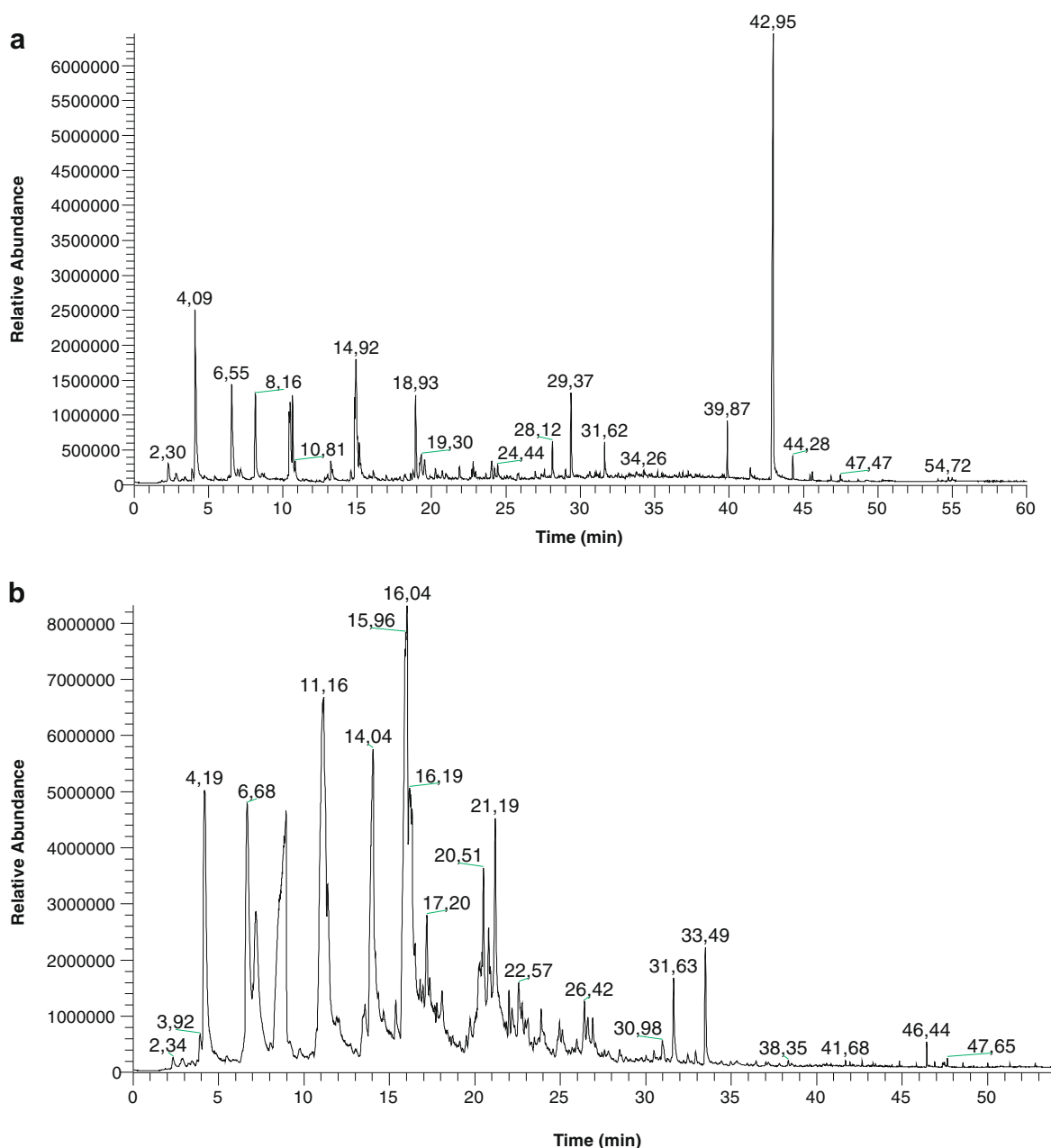


Fig. 2. Typical HS-SPME-GC-MS chromatograms of roasted and ground coffee using different SPME fibers: (a) PDMS, (b) CAR/PDMS, (c) PDMS/DVB, and (d) CAR/DVB/PDMS. Peak identification corresponds to retention time values.

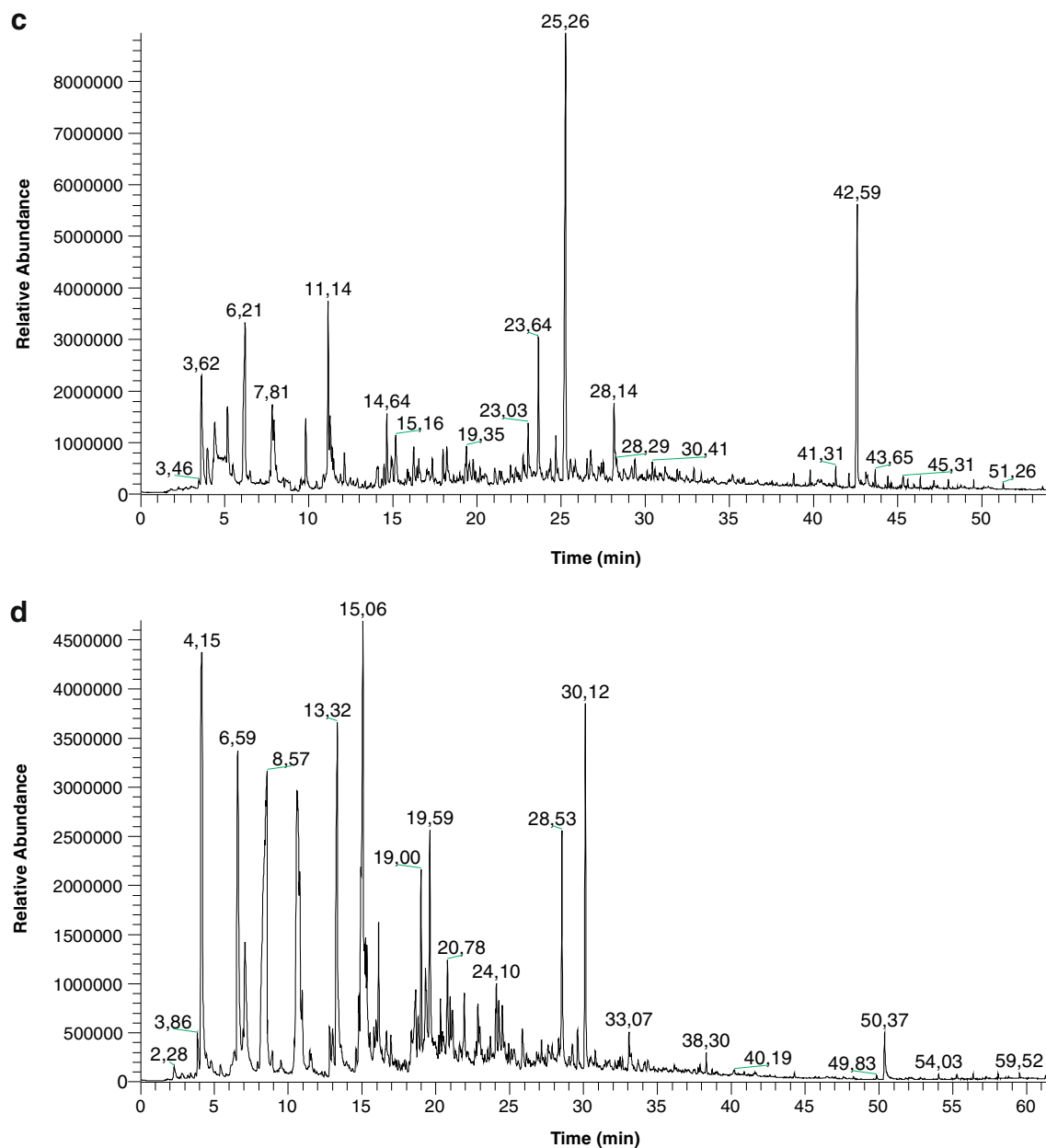


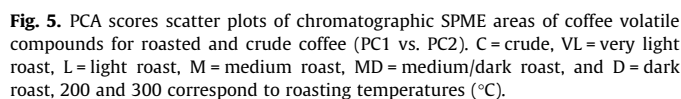
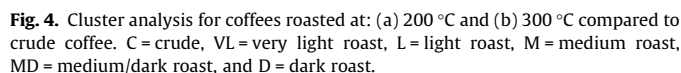
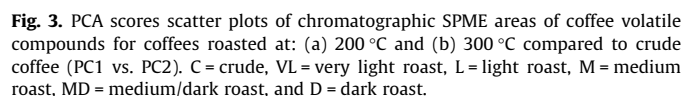
Fig. 2 (continued)

139 substances were detected employing PDMS, CAR/PDMS, PDMS/DVB, and DVB/CAR/PDMS fibers, respectively. Even though the total amount of extracted substances was similar for the PDMS/DVB and DVB/CAR/PDMS fibers, peak intensity was higher for the DVB/CAR/PDMS fiber. Based on these results, the triple phase fiber (DVB/CAR/PDMS) was selected to conduct the remaining tests. Results are qualitatively similar to those of Rojas Monroy (2005), who compared coffee headspace SPME extraction with both PDMS and DVB/CAR/PDMS fibers. This author reported extraction of 76 and 33 substances employing the DVB/CAR/PDMS and PDMS fibers, respectively.

A total of 86 peaks were detected in the headspace of raw coffee. Given the extensive amount of substances detected and the fact that an extensive characterization of the volatile profiles of both raw and roasted coffees is available in the literature (Czerny and Grosch, 2000; Sanz et al., 2001; Rocha et al., 2003; Mondello et al., 2004; Zambonin et al., 2005), discussion will be mainly focused on specific substances that are of interest in terms of roast-

ing degree assessment and differentiation. Among the detected peaks, 48 were tentatively identified, including benzenic and phenolic compounds, alkenes, esters, alcohols, and ketones. Regarding pyridines, pirazines, furans, and thiazoles, only one compound of each class was detected in raw coffee. Approximately 30% of the substances detected in crude coffee were also detected in at least one of the evaluated roasting degrees. Among the substances that were present only in the headspace of raw coffee, 3-isobutyl-2-methoxypyrazine has been reported in the literature as a potent odorant responsible for the characteristic, peasy odor note of raw coffee (Czerny and Grosch, 2000). Other compounds previously listed as potent odorants of raw coffee such as 4-ethylguaiacol, vanillin, and other pyrazines were not detected in this study, probably due to the fact that such compounds have been previously reported in much lower concentrations than 3-isobutyl-2-methoxypyrazine, and so could present *S/N* ratio below the limit established in this study. It is noteworthy to mention that such compounds have been known to present much lower odor activity

A wide variety of compounds were detected in roasted coffee, with the number of substances increasing with both roasting degree and temperature. These include furans, ketones, pyrazines, pyridines, and pyrroles that have been previously reported in the literature (De Maria et al., 1996; De Maria and Moreira, 1999, 2000; Mayer and Grosch, 2001; Grosch, 2001; Sanz et al., 2001; Costa Freitas and Mosca, 1999; Lee and Shibamoto, 2002; Mondello et al., 2004; Rocha et al., 2003; Yeretizian et al., 2003; Zambonin et al., 2005). The following substances, tentatively identified by comparison with mass spectra library, to our best knowledge, have not been reported by previous studies to be present in roasted coffee headspace profiles: 1,3,5,7-cyclooxatetraene, 2-carboxaldehyde-1-methyl-1H-pyrrol, 1-(2-furanyl)-1-propanone, 1-(2-piridinyl)-ethanone, 2,5-furandione, 1,2,4-triazol [4,3a] pyridine, (1-methyl-1H-pyrrol-2yl)-1-ethanone, 1-carboxylic acid-2-ciclopentene, 2-(2-furanyl)methyl-5-methylfuran, 3-phenylfuran, pentadenoic acid, 2-furanylmethylester, 2-hydroxy-1,7-naftiridine, 4-methylpyrrol[1.2a]pyrazine, 2,2-[oxybis(methylene)]bis-furan, 4methylpyrrol-[1.2a]pyrazine, 4-methyl-2(1H)quinolinone, 4-ethenylbenzene, 1,2-dimethoxy-2,5-dihydroxypropiophenone, 2-[1,3-butadienyl]1,3,5-trimethyl-benzene, α -Furfuryliden- α -furylmethylamine, hidroxytoluene butilate, *n*-nonylphenol, hexadenoic acid-mono (2-ethylhexil) ester, hexadenoic acid-methyl ester, 1-methyl-piperidinemethanol. Since these compounds were not reported before, their identity confirmation should be pursued in further studies as newly unconfirmed compounds in coffee head-

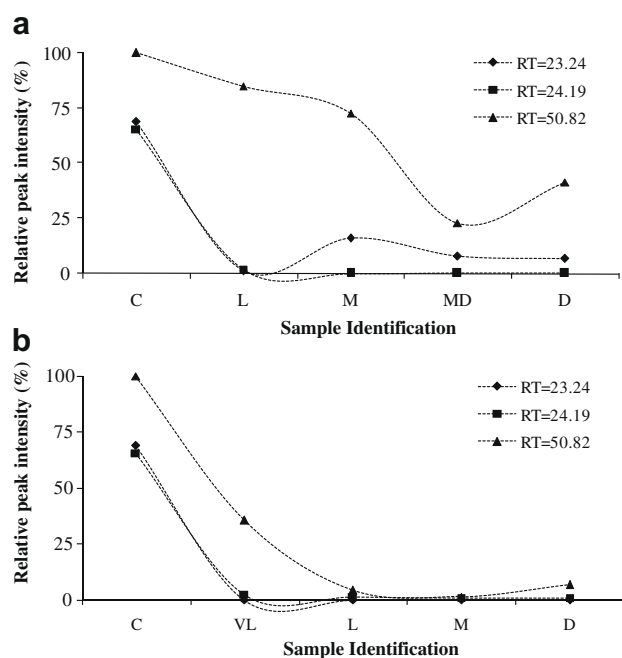


Given the extensive number of compounds that were detected in all roasted coffee samples, regardless of the degree of roast, multivariate statistical analysis (PCA and HCA) was performed in order to verify the possibility of discrimination between roasting degrees and roasting temperatures based on the volatile profiles. The central idea of Principal Component Analysis (PCA) is to reduce the

Table 2

Tentative identification of aroma components with the most pronounced effect on PC1 and PC2 values.

Retention time (min)	S/N ^a	Compound	m/z of the most intense ions (relative abundance %)
<i>Crude coffee</i>			
23.24	525	Unidentified	69 (100), 55(72), 67(70)
24.19	219	Methyl salicylate	120 (100), 92 (90), 152 (50)
50.82	2003	Caffeine	194 (100), 55 (20), 109 (10)
<i>Roasted Coffee – 200 °C</i>			
4.25	5410	Unidentified	80 (100), 79(36), 130(25)
6.69	1935	2-Methyl pyrimidine	94 (100), 67 (40), 93 (15)
7.16	707	Furfural	95 (100), 96 (30), 97 (10)
8.50	906	3-Furanmethanol	81 (100), 98 (55), 97 (50)
13.46	1426	5-Methyl-2-furancarboxaldehyde	109 (100), 110 (50), 53 (30)
15.67	3297	2-Furanmethanol,acetate	98 (100), 81 (85), 140 (40)
19.14	1520	2,6-Diethyl pyrazine	135 (100), 136 (50), 107 (20)
30.32	2088	4-Vinylguaiaicol	150 (100), 135 (65), 107 (45)
<i>Roasted Coffee – 300 °C</i>			
4.13	7873	Pyridine	79 (100), 80 (85), 52 (50)
7.16	2970	Furfural	95 (100), 96 (30), 97 (10)
8.50	2086	3-Furanmethanol	81 (100), 98 (55), 97 (50)
13.46	3998	5-Methyl-2-furancarboxaldehyde	109 (100), 110 (50), 53 (30)
15.12	62	2-(<i>n</i> -Propyl)-pyrazine	94 (100), 107 (30), 58 (20)
30.32	4514	4-Vinylguaiaicol	150 (100), 135 (65), 107 (45)

^a Signal-to-noise ratio (average values including different roasting degrees).**Fig. 6.** Effect of roasting on peak relative intensity for the discriminating compounds of crude coffee: roasting at (a) 200 °C and (b) 300 °C (RT = retention time in minutes; C = crude, VL = very light roast, L = light roast, M = medium roast, MD = medium/dark roast, and D = dark roast).

dimensionality of a data set, explaining the variance structure. This is achieved by linear transformation of the original data set of variables into a smaller number of uncorrelated significant principal components (PC). Geometrically, this transformation represents the rotation of the original coordinate system and the direction of the maximum residual variance is given by the first principal component axis. The second principal component, orthogonal to the first one, has the second maximum variance and so on. In this way, projections preserving maximum amounts of statistical information can be visualized in order to display a more detailed study of the data structure. Hierarchical Cluster Analysis (HCA), on the other hand, is a qualitative pattern classification method based on creating tree-structured clusters (dendrogram) of data objects

(samples), according to distances between their profiles. HCA begins by finding the closest pair of cases and combines them to form a cluster. The algorithm proceeds one step at a time, joining pairs of cases, pairs of clusters or a case with a cluster, until all the cases are in one cluster. The steps are displayed as a dendrogram which represents the similarity groupings between objects (samples). The method is hierarchical because once two cases are joined in a cluster they remain joined. The K-means nearest group method was used to generate the dendrogram. The K-means algorithm assigns each point to the cluster whose center (also called centroid) is nearest. The center is the average of all the points in the cluster, i.e., its coordinates are the arithmetic mean for each dimension separately over all the points in the cluster.

PCA and HCA results are presented in Figs. 3–5. The biplots of the samples submitted to roasting at 200 °C and 300 °C are displayed in Fig. 3a and b, respectively. The data presented in Fig. 3 were based on: (a) 10 × 249 and (b) 10 × 269 data matrices comprising two crude and eight roasted coffee samples each. Regarding the volatile profiles of coffee roasted at 200 °C in comparison to crude coffee, the first two principal components (PCs) explained 48% and 36% of the chromatographic variance, respectively. For coffee roasted at 300 °C, the first component explained 74% of the chromatographic variance, whereas the second component accounted for 20%. For both roasting temperatures a clear separation between raw and roasted coffee could be obtained based on the first component alone, with such separation becoming more evident with the increase in roasting temperature. In both cases, raw coffee presented negative PC1 whereas roasted coffee presented positive PC1. There is also a clear separation of roasted coffee in two groups, with the samples submitted to light roasts being separated from the ones submitted to medium and dark roasts. Sample grouping was confirmed by Hierarchical Cluster Analysis (HCA), as can be seen in Fig. 4. Raw coffee samples were well separated from the roasted ones (0% similarity), confirming the significant effect of roasting on the coffee volatile profiles.

PCA analysis of all samples is displayed in Fig. 5, based on an 18 × 340 data matrix comprising two crude and 16 roasted coffee samples, with the first two principal components (PCs) accounting for 42% and 26% of the chromatographic variance, respectively. It is clear that roasting temperature presents a significant effect on the volatiles profile, promoting a separation of the roasted coffee samples into two groups by PC1 values. Coffees roasted at 200 °C pre-

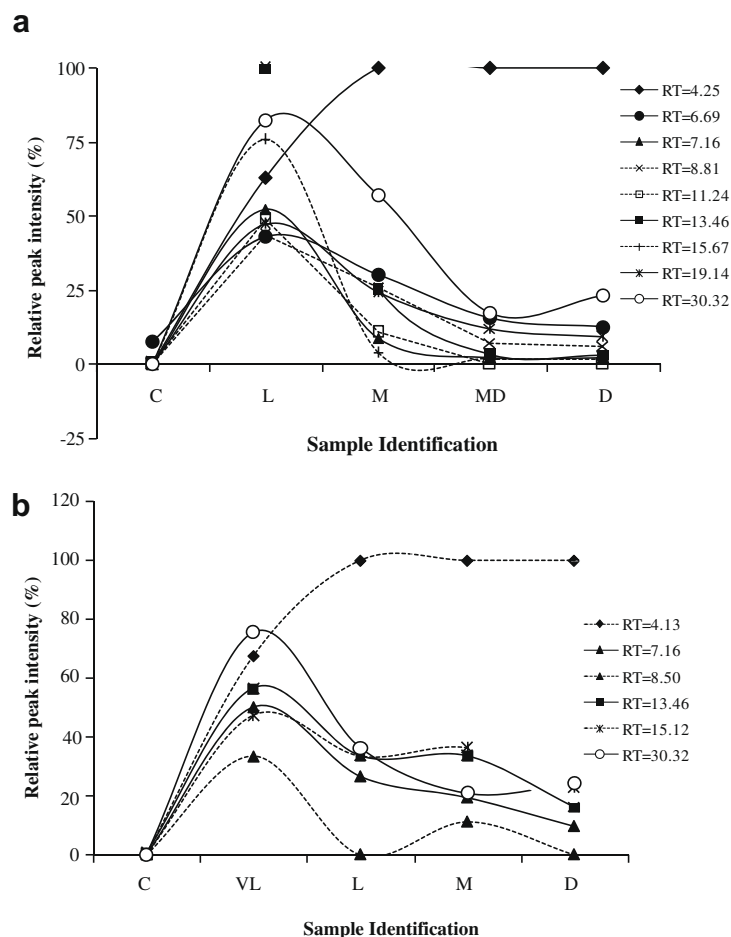


Fig. 7. Effect of roasting on peak relative intensity for the discriminating compounds of roasted coffee: roasting at (a) 200 °C and (b) 300 °C (RT = retention time in minutes; C = crude, VL = very light roast, L = light roast, M = medium roast, MD = medium/dark roast, and D = dark roast).

sented positive PC1 whereas coffee roasted at 300 °C presented negative PC1. PC2 values provided separation between crude and roasted coffee. Also, PC2 values decreased with an increase in roasting time, regardless of roasting temperature. The substances that presented higher influence on PC1 and PC2 are displayed in Table 2. Even though caffeine is known to be present in all coffee samples, regardless of roasting, the fact that such compound allowed for separation between crude and roasted coffees can be attributed to two factors: (i) the relative concentration of other volatile components with respect to caffeine being lower for crude than for roasted coffee and (ii) the fact that even though caffeine is not consumed during roasting, it has been previously reported to be dragged by the water vapor present in the exhaust gas as it evolves from the coffee beans, thus presenting lower concentrations for darker roasts (Dutra et al., 2001). The unidentified compound detected at 4.25 min, characteristic of coffee roasted at 200 °C, has been pointed in a previous study (Mancha Agresti et al., 2008) as a characteristic compound of healthy and sour coffee beans. The other substances listed in Table 2 as characteristic of roasted coffees have been identified by previous studies in the headspace of roasted coffee (De Maria et al., 1996; Sanz et al., 2001; Rocha et al., 2003; Mondello et al., 2004; Zambonin et al., 2005). Among the substances affecting the principal component in the roasted coffee samples, furfural, 3-furanmethanol, 5-methyl-2-furancarboxaldehyde and 4-vinylguaiacol (2-methoxy-4-vinylphenol) were detected for both roasting temperatures. However, the amount of these substances increased with roasting temperature, as can be verified by signal-to-noise ratio values dis-

played in Table 2. 4-Vinylguaiacol has been previously reported as a potent odorant of roasted coffee, being associated with a smoky aroma (Grosch, 2001). Pyridine was identified as the substance with the highest influence on both PC values for coffee roasted at 300 °C, being characteristic of samples submitted to a dark roast. Pyridines are formed due to trigonelline degradation, protein pyrolysis, Strecker degradation, and Maillard reactions (De Maria et al., 1996). Higher concentrations of pyridines have been reported for Arabica coffee samples in comparison to Robusta ones, under the same roasting conditions. Also, the amount of pyridines depends on the degree of roast, and such compounds have been reported to be characteristic of darker roasts (Dart and Nursten, 1985).

The relative peak intensities for the crude coffee discriminating compounds (Table 2) are plotted at varying degrees of roast in Fig. 6a and b, for 200 °C and 300 °C roasting temperatures, respectively. Compounds with retention times (RTs) 23.24 and 24.19 min presented similar behaviors for both temperature conditions, with a sharp decrease in relative intensity from crude to light roast degree. The decrease in relative intensity approaches zero and can be attributed to complete removal by volatilization, since their loss is related to the drying section of weight loss curves in Fig. 1a and b. Caffeine (RT = 50.82), however, presented distinct behavior for 200 °C and 300 °C roasting conditions. It presented a sharper decrease in intensity for roasting at 300 °C than it did at 200 °C, for lighter roasting degrees. Based on the roasting temperatures and on the weight loss curves in Fig. 1, one can infer that, for roasting at 300 °C, there probably were two competing mechanisms for caffeine removal during roasting (thus, decreasing its content in the

beans), water vapor drag and sublimation; whereas, for 200 °C, there was only water vapor drag, since the beans average temperature was surely borderline below that of caffeine sublimation.

The relative peak intensities for roasted coffee discriminating compounds (Table 2) at varying degrees of roast are plotted in Fig. 7a and b, for 200 °C and 300 °C roasting temperatures, respectively. For roasting at 200 °C, all the discriminating compounds presented similar behavior, attaining a maximum relative intensity at the lightest degree of roast, except for the unidentified compound, which increased its relative intensity as roasting progressed. The same behavior was observed for the discriminating compounds when roasting at 300 °C, except for pyridine (RT = 4.13) which also increased its relative intensity in the headspace profile as the degree of roast was intensified. Comparing the behavior of 4-vinylguaiacol (RT = 30.32) at different roasting conditions, it can be inferred that its formation and removal pathways are independent of the roasting process temperature, since it presented the same behavior for both 200 °C and 300 °C roasting temperatures. The behavior of furan and furanyl compounds (RTs = 7.16, 8.50, 13.46, and 15.67) at varying degrees of roast observed in this study corroborated the dependence on roasting observed by Liardon and Ott (1984). Also, it can be inferred that these classes of compounds, resulting from the thermal degradation of carbohydrates, are mostly formed at the onset of pyrolysis, given that they presented the maximum relative intensity at the lightest roasting degree.

It is noteworthy to point out that samples that were considered to present the same degree of roast, on the basis of color, and weight loss, where clearly separated from each other. Thus, the results presented in this study show that such parameters are not reliable as roasting degree assessment criteria and that roasting temperature must also be taken into account.

4. Conclusions

The characterization of headspace volatile profiles of coffee under two different processing temperatures was successfully carried out by multivariate statistical analysis of the respective SPME/GC–MS chromatograms. The analysis allowed for the discrimination between crude and roasted samples, and between samples roasted to varying degrees under different temperatures, regardless of compound identification. The behavior of the discriminating compounds was correlated to the weight loss curves during roasting, which in turn presented distinct behavior for different roasting temperatures. Caffeine, although not volatile, was encountered in all headspace profiles and was one of the discriminating compounds for crude coffee. Pyridine was the main discriminating compound for coffees roasted at high temperatures and more intense degrees of roast. Although discrimination of coffee samples roasted to varying degrees was achieved by analyzing the effect of hundreds of compounds altogether, the compounds that presented the most pronounced effect on the discrimination of roasting degrees and temperatures were those generated right after the onset of pyrolysis. The results presented in this study show that color and weight loss alone are not reliable as roasting degree assessment criteria and that roasting temperature must also be taken into account.

Acknowledgements

The authors acknowledge financial support from the following Brazilian Government Agencies: CAPES, CNPq and FAPEMIG.

References

Clarke, R.J., 1987. Roasting and Grinding. In: Clarke, R.J., Macrae, R. (Eds.), *Coffee*, vol. 2: Technology. Elsevier Applied Science, London, pp. 73–108.

- Costa Freitas, A.M., Mosca, A.I., 1999. Coffee geographic origin – an aid to coffee differentiation. *Food Research International* 32 (8), 565–573.
- Cristo, H.P., Martins, M.A., Oliveira, L.S., Franca, A.S., 2006. Transverse flow of coffee beans in rotating roasters. *Journal of Food Engineering* 75, 142–148.
- Czerny, M., Grosch, W., 2000. Potent odorants of raw arabica coffee. Their changes during roasting. *Journal of Agricultural and Food Chemistry* 48 (3), 868–872.
- Dart, S.K., Nursten, H.E., 1985. Volatile components. In: Clarke, R.J., Macrae, R. (Eds.), *Coffee*, vol. 1: Chemistry. Elsevier Applied Science, London, pp. 223–265.
- De Maria, C.A.B., Trugo, L.C., Aquino Neto, F.R., Moreira, R.F.A., Alviano, C.S., 1996. Composition of green coffee water-soluble fractions and identification of volatiles formed during roasting. *Food Chemistry* 55 (3), 203–207.
- De Maria, C.A.B., Moreira, R.F.A., Trugo, L.C., 1999. Componentes voláteis do café torrado. Parte I: Compostos Heterocíclicos (Volatile components in roasted coffee. Part I: heterocyclic compounds), *Química nova* 22 (2), 209–217 (in Portuguese).
- De Maria, C.A.B., Moreira, R.F.A., Trugo, L.C., 2000. Componentes voláteis do café torrado. Parte II: Compostos alifáticos, alicíclicos e aromáticos (Volatile components in roasted coffee. Part II: Aliphatic, alicyclic and aromatic compounds), *Química nova* 23(2), 195–203 (in Portuguese).
- Dutra, E.R., Oliveira, L.S., Franca, A.S., Ferraz, V.P., Afonso, R.J.C.F., 2001. A preliminary study on the feasibility of using the composition of coffee roasting exhaust gas for the determination of the degree of roast. *Journal of Food Engineering* 47 (3), 241–246.
- Franca, A.S., Oliveira, L.S., 2008. Chemistry of defective coffee beans. In: Koeffer, E.N. (Ed.), *Food Chemistry Research Developments*. Nova Publishers, New York, p. 34p. Chapter 4.
- Franca, A.S., Oliveira, L.S., Mendonça, J.C.F., Silva, X.A., 2005. Physical and chemical attributes of defective crude and roasted coffee beans. *Food Chemistry* 90, 84–89.
- Grosch, W., 2001. Chemistry III: volatile compounds. In: Clarke, R.J., Vitzthum, O.Z. (Eds.), *Coffee: Recent Developments*. Blackwell Science, Oxford, pp. 68–89.
- Lee, K.-G., Shibamoto, T., 2002. Analysis of volatile components isolated from Hawaiian green coffee beans (*Coffea arabica* L.). *Flavour and Fragrance Journal* 17 (5), 349–351.
- Liardon, R., Ott, U., 1984. Application of multivariate statistics for the classification of coffee headspace profiles. *Lebensmittel-Wissenschaft und-Technologie* 17 (1), 32–38.
- Mancha Agresti, P.C., Franca, A.S., Oliveira, L.S., Augusti, R., 2008. Discrimination between defective and non-defective Brazilian coffee beans by their volatile profile. *Food Chemistry* 106 (2), 787–796.
- Mayer, F., Grosch, W., 2001. Aroma simulation on the basis of the odorant composition of roasted coffee headspace. *Flavour and Fragrance Journal* 16 (3), 180–190.
- Mondello, L., Casilli, A., Tranchida, P.Q., Dugo, P., Costa, R., Festa, S., Dugo, G., 2004. Comprehensive multidimensional GC for the characterization of roasted coffee beans. *Journal of Separation Science* 27 (5–6), 442–450.
- Oliveira, S.D., Franca, A.S., Glória, M.B.A., Borges, M.L.A., 2005. The effect of roasting on the presence of bioactive amines in coffees of different qualities. *Food Chemistry* 90 (1–2), 287–291.
- Purdon, M.P., McCamey, D.A., 1987. Use of a 5-caffeoylquinic acid/caffeine ratio to monitor the coffee roasting process. *Journal of Food Science* 52 (6), 1680–1683.
- Rocha, S., Maeztu, L., Barros, A., Cid, C., Coimbra, M.A., 2003. Screening and distinction of coffee brews based on headspace solid phase microextraction/gas chromatography/principal component analysis. *Journal of the Science of Food and Agriculture* 84, 43–51.
- Rojas Monroy, G.M.E., 2005. Caracterización del aroma del café molido de Puerto Rico, mediante la técnica de micro extracción en fase sólida (SPME) y cromatografía de gas acoplada a espectrometría de masas (CG/MS) (Characterization of ground coffee aroma from Puerto Rico using the solid phase microextraction technique (SPME) and gas chromatography coupled to mass spectrometry (GC/MS)). Universidad de Puerto Rico Recinto Universitario de Mayaguez. MS Thesis, Food Science, 156p.
- Ryan, D., Shellie, R., Tranchida, P., Casilli, A., Mondello, L., Marriott, P., 2004. Analysis of roasted coffee bean volatiles by using comprehensive two-dimensional gas chromatography–time-of-flight mass spectrometry. *Journal of Chromatography A* 1054 (1–2), 57–65.
- Sanz, C., Ansorena, D., Bello, J., Cid, C., 2001. Optimizing headspace temperature and time sampling for identification of volatile compounds in ground roasted arabica coffee. *Journal of Agricultural and Food Chemistry* 49, 1364–1369.
- Sivetz, M., Desrosier, N.W., 1979. *Coffee technology*. AVI Publishing, Westport.
- Vasconcelos, A.L., 2005. Caracterização de grãos defeituosos de café quanto a aspectos físico-químicos e teores de amins, para diversos graus de torração (Characterization of defective coffee beans with respect to physical and chemical attributes and amine levels, for several roasting degrees). Universidade Federal de Minas. MS Thesis, Food Science, 101p.
- Yeretzian, C., Jordan, A., Lindinger, W., 2003. Analysing the headspace of coffee by proton-transfer-reaction mass-spectrometry. *International Journal of Mass Spectrometry* 223–224, 115–139.
- Zamboni, C.G., Balest, L., De Benedetto, G.E., Palmisano, F., 2005. Solid-phase microextraction-gas chromatography mass spectrometry and multivariate analysis for the characterization of roasted coffees. *Talanta* 66 (1), 261–265.